

Integrity assurance of Kalman-filter based GNSS/IMU integrated systems against IMU faults for UAV applications

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ABSTRACT

This study proposes an integrity architecture for an Extended Kalman filter (EKF) based Global Navigation Satellite System (GNSS)/Inertial Measurement Unit (IMU) integrated system to assure integrity of unmanned aerial vehicle (UAV) navigation systems. An integrity risk allocation tree is developed for each sensor fault hypothesis including the nominal hypothesis, a GNSS fault hypothesis, and an IMU sensor fault hypothesis. This paper then proposes a real-time EKF vertical protection level (VPL) against IMU sensor faults to assure navigation integrity based on the relationship between EKF innovations and EKF state errors resulting from potential IMU faults and measurement noise. Simulations for the derived EKF VPLs are conducted under a specific IMU-fault condition and under no-IMU-fault condition to investigate typical performance. This study will be extended to assure the navigation integrity of different types of multi-sensor systems for UAV applications.

1.0 INTRODUCTION

As many beneficial civilian applications of unmanned aerial vehicles (UAVs) have been proposed, the need for an infrastructure that enables the widespread use of UAVs in low-altitude airspace has emerged. Accordingly, the concept of an unmanned aircraft system (UAS) traffic management (UTM) system is being developed to enable safe and efficient low-altitude airspace operations [1]. A desired attribute of the UTM system is that it would not require human operators to monitor every vehicle continuously [1]. This approach would need to ensure safety under all UAV operational conditions, including the navigation status, surrounding traffic, weather conditions, etc. An essential component for ensuring safety and self-protection is a high-integrity UAV navigation system that can detect navigation faults in real time as well as provide integrity risk information to users, such as the protection levels bounding errors in the estimated position solutions to the required probability of hazardous misleading information (HMI).

Over the past three decades, high-integrity GNSS-based navigation systems for civilian aviation have been extensively developed, including the Satellite-Based Augmentation System (SBAS) [2], the Ground-Based Augmentation System (GBAS) [3], and applications of Receiver Autonomous Integrity Monitoring (RAIM) [4]. These systems provide the level of integrity that is required for each flight procedure. In addition to GNSS, civil aircraft utilize other sensors, such as the outputs of an inertial measurement unit (IMU) and barometric altitude. Unlike GNSS integrity techniques, integrity assurance for these sensors primarily relies on hardware redundancy [5]. However, despite the hardware-redundant setup, recent civil aircraft accidents have indicated that sensor malfunctions can result in critical failures [6]. One example of an IMU failure was reported in the investigation of Qantas Airbus A330-303 Flight QF72 on 7 October 2008. During this flight, it was found that one of the air data inertial reference units (ADIRUs) was providing erroneous data on many parameters to other aircraft systems. This led to a flight upset and 11 seriously-injured passengers [6].

As the importance of UAV navigation integrity increases, many studies have been conducted to assure GNSS-based UAV navigation integrity comparable to the integrity requirements used for civil aviation [7, 8]. However, those studies are limited to providing assurance for GNSS-based navigation systems. To assure the integrity of systems that combine GNSS with other sensors, the integrity impacts of all sensor faults must be considered.

An IMU is a typical sensor integrated with GNSS for UAV navigation to provide better accuracy, continuity, and reliability of navigation solutions as well as attitude with higher update rates. However, due to a lack of

knowledge in the reliability of IMU sensors, utilizing low-cost IMU sensors has limitations to safety-critical applications. A previous study stated that IMUs are more prone to faults under highly vibrating environments in which an UAS usually operates [9]. Another previous paper listed fault types of IMU sensors, including a step fault, a ramp fault and a random noise fault. The paper also summarized potential causes for each of those fault types [10].

One method for assuring IMU sensor integrity in a UAV navigation system can be the utilization of multiple redundant IMUs as in civil aviation. Such a system can continue operations by detecting individual IMU sensor faults and switching to backup IMU sensors. However, providing enough redundant equipment to make this possible is undesirable for low-cost UAVs with tight mass and power constraints. Many studies on fault detection algorithms have been conducted [11], [12]. However, most of proposed algorithms focused on quantifying the fault-detection capability, without rigorous quantification of fault impacts to user navigation estimates and fault monitors. In safety-critical applications, beyond fault detection capability, assuring the navigation integrity risk against sensor faults and providing the information to operating users are essential.

This study targets low-cost UAV applications and proposes methods to assure total navigation system integrity of a GNSS/IMU integrated system without having redundant IMUs by developing an integrity architecture for a GNSS/IMU integrated navigation system and computing real-time vertical protection levels (VPLs) for each nominal, GNSS, and IMU fault hypothesis. VPLs of the integrated system can provide a primary layer of protection for user navigation solution with a required level of system integrity, before monitors detect faults. In this study, GNSS and a single IMU sensor are integrated with an Extended Kalman filter (EKF), which is the most widely used algorithm for sensor integration [13], [14]. An EKF is a recursive estimator that utilizes past measurements to generate the current state estimates, unlike a least-square (LS) filter that is a snapshot estimator. Hence, position estimates as well as sensor fault impacts recursively propagate through EKF algorithms. This recursive process should be represented within fault monitoring algorithms and in the formulation of the VPL equations.

As a response, several recent studies have proposed methods to evaluate the integrity risk of KF estimates against faults in a measurement sequence (i.e., GNSS measurement faults) which are applied in a measurement update step in KF [15]-[17]. The published methods to quantify the integrity risk in a sequential KF in [15]-[17] can be classified into two approaches. Studies in [15] and [16] both quantified the worst-case integrity risk by determining the worst-case fault vector analytically based on the relationship between state error and range residual based monitor test statistics formulated using a batch least square expression in [15], and a recursive expression in [16]. [16] specifically applied the KF based integrity evaluation algorithm to a GNSS spoofing detection in a GNSS/INS integrated system by assuming that the INS is a fault-free sensor.

As another approach, [17] proposed computing protection level through a residual-based RAIM algorithm in real-time instead of computing the worst-case integrity risk. To improve computational efficiency, a snapshot Weighted Least Square (WLS) residual-based RAIM algorithm for fault monitoring was used while user position was estimated by the EKF. A Minimum Detectable Error (MDE) derived from the monitor was then converted to a range domain with an assumption of a single satellite fault. Since the MDE determined by the snapshot based fault monitor does not include recursive effects of the EKF, which a user is utilizing for positioning, a user mean position bias was recursively updated applying the MDE and KF parameters and is applied to a VPL equation.

The primary focus of this paper is to assure the system integrity of EKF-based GNSS/IMU systems by developing algorithms for assuring integrity against IMU sensor faults. As IMU sensor measurements are used in system propagation steps in the KF, the previous studies cited above ([15]-[17]) cannot be directly applied to evaluate integrity risks against IMU faults. Instead, this study extends the concepts of previous work on KF integrity evaluations within a newly designed novel integrity architecture for a GNSS/IMU integrated system. The architecture exploits existing GNSS integrity methods to guarantee the required levels of integrity for GNSS. Based on the monitored GNSS measurements, potential IMU sensor faults are monitored using the EKF innovations under the assumption of no GNSS fault. Unlike the measurement sequence fault case, it is possible to fully capture the fault impact on the user position state in real-time using an EKF innovation vector under IMU fault conditions. Exploiting this fact, a VPL is derived utilizing the EKF innovation with additional uncertainty noise bounding terms which bound a potential user position with a level of system integrity.

In Section 2.0, the newly proposed integrity architecture for a GNSS/IMU integrated navigation system is introduced. This architecture includes a fault tree to allocate integrity risk to each fault hypothesis along with derivations of VPL equations for each hypothesis. Section 3.1 provides the details of how IMU faults impact the

EKF states and analytically derives the resulting state errors. Section 3.2 introduces the derivation of a real-time VPL equation for the IMU fault hypothesis based on the EKF innovations. Simulation results for the performance of VPLs of the integrated navigation system under an IMU fault condition, and under nominal conditions are shown in Section 4.0 and in Section 5.0, respectively. Finally, conclusions are given in Section 6.0.

2.0 INTEGRITY ARCHITECTURE OF A GNSS/IMU INTEGRATED NAVIGATION SYSTEM

This study proposes an integrity architecture for a GNSS/IMU integrated navigation system that requires a high level of system integrity. Fig. 1 illustrates an overview of the architecture. The flowchart on the left in Fig. 1 shows a tightly coupled EKF-based GNSS/IMU integration algorithm. An IMU sensor is used to propagate the system, and GNSS pseudoranges are used to update in a measurement update step. In the integration algorithm, integrity monitoring algorithms are included for both a GNSS system and an IMU sensor. The architecture exploits the existing GNSS integrity systems to guarantee the required levels of integrity against GNSS failures. In this study, the WLS-RAIM algorithm, which is the most typical and computationally efficient form of integrity monitoring, is applied for a GNSS integrity monitoring algorithm. In addition to the GNSS system failure, this study newly proposes an integrity assurance against IMU sensor failures to fully assure the total navigation system integrity of the integrated system. The total navigation system integrity is assured by computing the EKF VPLs for each fault hypothesis, as shown on the right side of Fig. 1, and by comparing it with a pre-defined threshold during operations. Section 2.1 introduces an algorithm of a tightly coupled EKF for GNSS/IMU integrations. Subsequently, Section 2.2 introduces an integrity tree that includes an allocation for a newly suggested IMU fault hypothesis for the integration system and the corresponding EKF VPLs for each hypothesis.

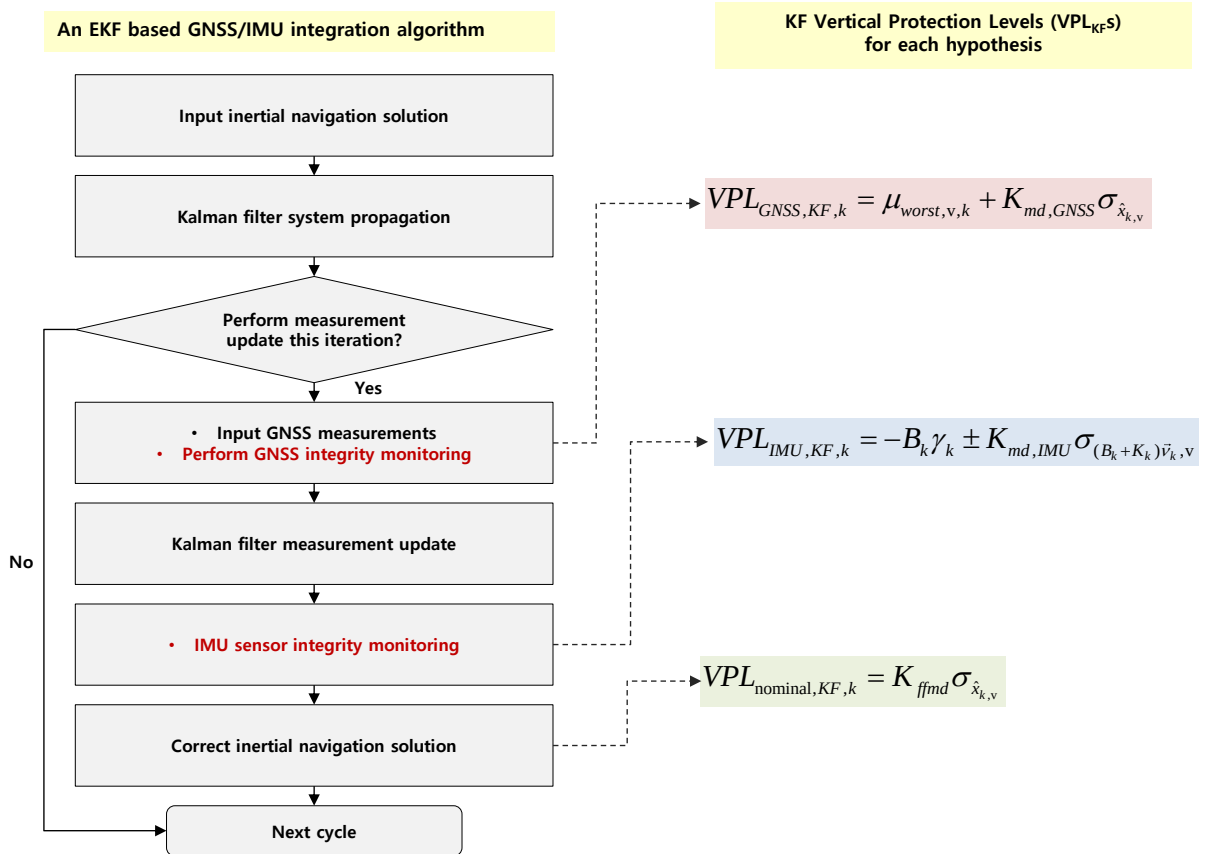


Fig. 1. EKF-based GNSS/IMU integration algorithm and VPL equations for each fault hypothesis

2.1 Tightly-coupled EKF for GNSS/IMU integration

In this study, a form of direct EKF is established in which total quantities such as position, velocity, and attitude are estimated directly by the EKF state vector. A state vector includes the position, velocity, attitude, inertial sensor biases, receiver clock offset, and receiver clock drift. The EKF state equations of a linear dynamic system at any discrete time k are described by a system propagation Equation (1) and a measurement update equation (2).

The IMU sensor measurements are considered as a control input $\vec{u}_k = [\vec{f}_{ib,meas}^b; \vec{\omega}_{ib,meas}^b]$, which is a 6×1 vector (three-axis measurements for an accelerometer and a gyroscope) and used in the system propagation step. The GNSS pseudorange measurements are used as measurements in the measurement update step. The EKF states are recursively propagated through equations (1) and (2) for each time epoch [9, 10].

$$\bar{x}_k = \Phi_k \hat{x}_{k-1} + \Gamma_k \vec{u}_k + \Gamma_w \vec{w}_k \quad (1)$$

$$\hat{x}_k = \bar{x}_k + K_k (\bar{z}_k - H_k \bar{x}_k) \quad (2)$$

where

$\bar{x}_k$ is the predicted state vector at time k

$\hat{x}_k$ is the measurement updated state vector at time k

Φ_k is the state transition matrix

Γ_{uk} is the input coefficient matrix

Γ_{wk} is the noise coefficient matrix

$\vec{u}_k$ is the control input vector (IMU sensor measurements in this study)

$\vec{w}_k$ is the process noise vector

K_k is the Kalman gain matrix

$\bar{z}_k$ is the measurement vector (GNSS pseudorange measurements in this study)

H_k is the observation matrix

2.2 Integrity risk allocation tree and EKF vertical protection levels

To assure the integrity of the GNSS/IMU integrated system against all possible sensor fault hypothesis, an integrity risk allocation tree is developed as in Fig. 2. The total probability of HMI (P_{HMI}) is separated into P_{HMI} in the vertical domain ($P_{HMI,v}$) and P_{HMI} in the horizontal domain. Fig. 2 describes a design for the P_{HMI} in the vertical position domain as an example. In this study, three fault hypotheses are defined including a nominal hypothesis, a GNSS fault hypothesis, and an IMU sensor fault hypothesis. The total $P_{HMI,v}$ is allocated to each hypothesis, and the $P_{HMI,v}$ s for each hypothesis are used in the integrity monitors as well as the corresponding VPL computations with prior probabilities of GNSS and IMU sensors. The $P_{HMI,Total}$ is a system-dependent requirement, and the prior probabilities of sensors are hardware-dependent parameters. Sections 2.2.1 to 2.2.3 describe the VPLs for each hypothesis.

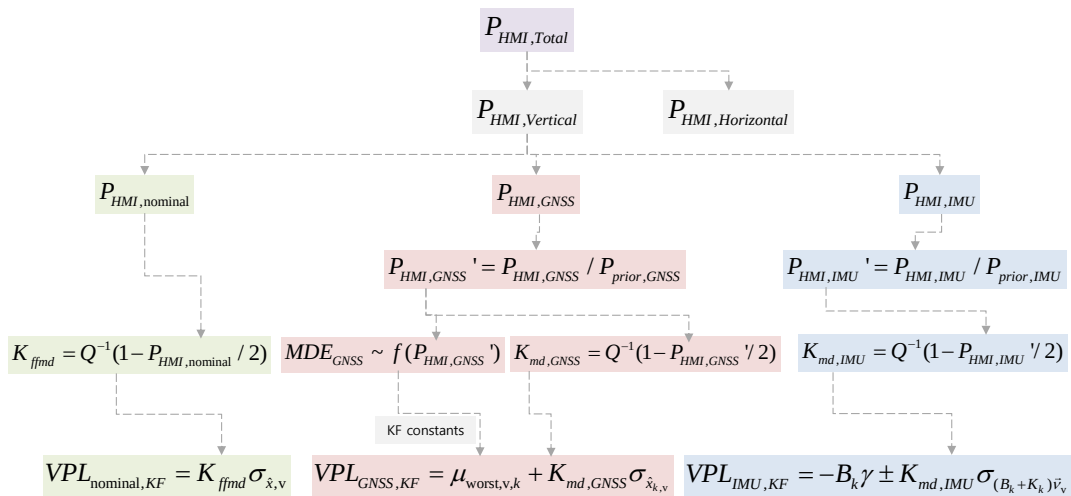


Fig. 2. Integrity risk allocation tree for a GNSS/IMU integrated system

2.2.1 VPL under GNSS fault hypothesis ($VPL_{GNSS,KF}$)

KF-based integrity evaluation algorithms against GNSS fault hypothesis have been proposed previously [15]-[17]. Among those studies, [17] proposed computing a real-time KF VPL against GNSS fault hypothesis based on a WLS RAIM algorithm. This study adopted the method for computing the KF VPL against GNSS faults, but extended it to a GNSS/IMU integration system. A test statistic is computed as a difference between GNSS pseudorange measurements and WLS filter predicted measurements. A MDE is determined by the non-centrality parameter λ_k of a chi-square distribution, which is obtained as the weighted sum of the squared errors (WSSE) of the test statistics.

The corresponding fault vector $f_{GNSS,i}$ is determined by projecting the MDE to a single i^{th} satellite by assuming a single satellite fault. (3) is the equation for the recursive propagation of a mean position bias μ_k against the i^{th} satellite fault at epoch k . The EKF parameters in (3) are identical with those introduced in (1) and (2).

$$\mu_{k,i} = (I - K_k H_k) \Phi_k \mu_{k-1,i} + K_k f_{GNSS,k,i} \quad (3)$$

Finally, $VPL_{GNSS,KF}$ is computed as the sum of the absolute value of the worst-case bias from all the potential satellites and a nominal sigma, determined by the EKF output variance as shown in (4), where $f_{\bar{z}_k, \text{worst}}$ is the worst fault vector determined from all satellites. For more details on computing the worst mean position bias, please refer to [17].

$$VPL_{GNSS,KF,k} = \mu_{\text{worst},v,k} + K_{md,GNSS} \sigma_{\hat{x}_{k,v}} \quad (4)$$

where $\mu_{\text{worst},v,k} = |(I - K_k H_k) \Phi_k \mu_{\text{worst},v,k-1}| + |K_k|_v f_{\bar{z}_k, \text{worst}}$

2.2.2 VPL under IMU fault hypothesis ($VPL_{IMU,KF}$)

This study newly proposes a real-time $VPL_{IMU,KF}$ under an IMU fault hypothesis by directly estimating a real-time position state bias estimate using an EKF innovation ($\vec{\gamma}_k$). $VPL_{IMU,KF}$ comprises a position state bias estimate term ($-B_{k,v} \vec{\gamma}_k$) and a noise bounding term ($K_{md,IMU} \sigma_{(B_k + K_k) \vec{v}_{k,v}}$) as in (5). Because the position state bias estimate includes uncertainties from the innovation noise, the noise bounding term is added to properly overbound the resulting position state errors to the level of integrity under the IMU fault conditions. Unlike $VPL_{GNSS,KF}$, this approach directly estimates the position state bias estimate even though noise is included in it. Thus, the absolute value on the bias estimate is not required to formulate the VPL. Instead, a choice is made to utilize two VPLs that have a positive sigma bound and a negative sigma bound from the bias estimate correspondingly. Having two VPLs would lower the conservatism compared to having an absolute value of the bias estimate on $VPL_{IMU,KF}$. To monitor excessive IMU faults that would threaten a user navigation system, the computed real-time $VPL_{IMU,KF}$ is continuously compared with a corresponding threshold or a vertical alert limit (VAL), which is defined for each system requirement. If a VPL_{IMU} exceeds those pre-defined thresholds, the system is declared unavailable.

Section 3.0 discusses more details on the derivation of $VPL_{IMU,KF}$.

$$VPL_{IMU,KF,k} = -B_{k,v} \vec{\gamma}_k \pm K_{md,IMU} \sigma_{(B_k + K_k) \vec{v}_{k,v}} \quad (5)$$

2.2.3 VPL under nominal conditions ($VPL_{nominal,KF}$)

A VPL under a nominal condition is defined as (6). The output variance $\sigma_{\hat{x}_{k,v}}$ in (6) from the EKF is used to compute the $VPL_{nominal,KF}$. To assure that the $VPL_{nominal,KF}$ reliably bounds the actual state errors, it is important to bound the process noise variance and the measurement noise variance, which are inputs to the EKF. (7) is an expression of the covariance matrix of a measurement updated state vector used for determining the $\sigma_{\hat{x}_{k,v}}$. To overbound the output covariance from an EKF, (8) and (9) should be satisfied where $\prec$ indicates ‘overbound’. The two equations indicate that an output sigma from an EKF is overbounded if an input process noise variance and an input measurement noise variance overbound an actual noise distribution.

$$VPL_{\text{nominal},KF,k} = K_{\text{ffmd}} \sigma_{\hat{x}_{k,v}} \quad (6)$$

$$P_k^+ = (I - K_k H_k) P_k^- \quad (7)$$

$$\begin{aligned} P_{k,true}^+ &\prec P_{k,bound}^+ \quad \text{if } (I - K_{k,true} H_k) \prec (I - K_{k,bound} H_k) \\ &\therefore K_{k,true} \succ K_{k,bound} \quad (8), \text{ and} \\ &\therefore R_{true} \prec R_{bound} \end{aligned}$$

$$\begin{aligned} P_{k,actual}^- &\prec P_{k,bound}^- \\ \therefore \Phi_k P_{k-1}^+ \Phi_k^T + \Gamma_w Q_{w,actual} \Gamma_w^T &\prec \Phi_k P_{k-1}^+ \Phi_k^T + \Gamma_w Q_{w,bound} \Gamma_w^T \quad (9) \\ \therefore Q_{w,actual} &\prec Q_{w,bound} \end{aligned}$$

For the IMU noise variance, care should be taken because the IMU noise is time-correlated. Several studies have been conducted to assure integrity in safety-critical applications in the presence of time-correlated noise [18]-[20]. These studies investigated the effects of a time constant in a first-order Gauss Markov process model used to model the correlated noise. The methods should be applied when actual IMU sensor data are used. In this study, we simplified simulations by generating both GNSS and IMU sensor noise were in simulations. Thus, EKF output variances were bounded by applying the exactly known noise parameters to the EKF. Noise parameters used for the simulation will be introduced in Section. 4.1.2.

3.0 IMU FAULT IMPACT ON KALMAN FILTER STATE AND ERROR BOUNDING

3.1 EKF state error estimation

This section analytically derives the EKF state error resulting from faulty IMU measurements. The derivation of the EKF state error is based on a recursive estimation process of the EKF. An expression of a faulty IMU measurement is first defined in Section 3.1.1.

3.1.1 Defining an expression of a faulty IMU measurement

To quantify the impact of a faulty IMU measurement on the EKF state estimates, an expression of a faulty IMU measurement is defined based on kinematic inertial navigation equations. (11) shows the inertial navigation equations in the Earth Centered Earth Fixed (ECEF) coordinate frame [14]. Based on the equations, attitude is updated using gyroscope sensor measurements $\vec{\omega}_{ib}^b$, and velocity is updated using the updated attitude and accelerometer sensor measurements $\vec{f}_{ib}^b$. Finally, position is determined by integrating the update velocity.

$$\begin{aligned} \frac{\partial \vec{q}_b^e}{\partial t} &= \frac{1}{2} q^* (0, C_b^e \vec{\omega}_{ib}^b - \vec{\omega}_{ie}^e) \\ \frac{\partial \vec{v}_{eb}^e}{\partial t} &= C_b^e \vec{f}_{ib}^b - (2\vec{\omega}_{ie}^e) \times \vec{v}_{eb}^e + \vec{g}^e \quad (11) \\ \frac{\partial \vec{r}_{eb}^e}{\partial t} &= \vec{v}_{eb}^e \end{aligned}$$

where

$\vec{q}_b^e$ is the quaternion vector for an attitude estimation

C_b^e is the coordinate transformation matrix from the body to Earth frame

ω_{ie}^e is the Earth rotation rate

$\vec{v}_{eb}^e$ is the velocity vector

$\vec{g}^e$ is the gravity vector in ECEF frame

$\vec{r}_{eb}^e$ is the position vector

If a fault occurs on an IMU sensor, the computed attitude, velocity, and position solutions are sequentially affected by the fault, resulting in position errors on the user position solutions from a navigation standpoint. The velocity update equation is used as an example to explain a fault vector defined on the EKF states. In a faulty IMU condition, the time derivative of the velocity equation can be expressed as (12) (apostrophe (') indicates faulty parameter) assuming faulty accelerometer measurements as well as a faulty attitude estimate due to the previous faulty gyroscope measurements.

$$\dot{\vec{v}}_{eb}^e{}' = C_b^e{}' \vec{f}_{ib}^b{}' - (2\vec{\omega}_{ie}^e) \times \vec{v}_{eb}^e + \vec{g}^e \quad (12)$$

The faulty transformation matrix $C_b^e{}'$ and the faulty measurement input $\vec{f}_{ib}^b{}'$ are expressed as the sum of a nominal term and an additional term (ΔC_b^e and $\vec{f}_{\tilde{ib}}^b$) due to faults as

$$C_b^e{}' = C_b^e + \Delta C_b^e \quad (13)$$

$$\vec{f}_{ib}^b{}' = \vec{f}_{ib}^b + \vec{f}_{\tilde{ib}}^b \quad (14)$$

where ΔC_b^e is a difference between the true transformation matrix and the faulty transformation matrix, and $\vec{f}_{\tilde{ib}}^b$ is an accelerometer fault vector.

Substituting (13) and (14) into (12), the velocity state becomes,

$$\begin{aligned} \dot{\vec{v}}_{eb}^e &= (C_b^e + \Delta C_b^e)(\vec{f}_{ib}^b + \vec{f}_{\tilde{ib}}^b) - (2\vec{\omega}_{ie}^e) \times \vec{v}_{eb}^e + \vec{g}^e \\ &= C_b^e \vec{f}_{ib}^b - (2\vec{\omega}_{ie}^e) \times \vec{v}_{eb}^e + \vec{g}^e + \underbrace{C_b^e \vec{f}_{\tilde{ib}}^b + \Delta C_b^e(\vec{f}_{ib}^b + \vec{f}_{\tilde{ib}}^b)}_{\vec{f}_{\dot{v}_{eb}}^e} \end{aligned} \quad (15)$$

where $\vec{f}_{\dot{v}_{eb}}^e$ is the resultant bias vector on a velocity state due to the IMU sensor faults. In an EKF, the acceleration $\vec{f}_{ib}^b$ is represented as (16), which includes an accelerometer measurement, a bias on an accelerometer, and a process noise.

$$\vec{f}_{ib}^b = \vec{f}_{ib,meas}^b - \vec{b}_{f_{ib}}^b - \vec{w}_{accel} \quad (16)$$

By substituting (16) into (15), (15) becomes a continuous form of an EKF state equation as a function of EKF states, a control input, a process noise, and additionally the state fault vector as in (17).

$$\underbrace{\dot{\vec{v}}_{eb}^e}_{\dot{\vec{x}}} = \underbrace{-C_b^e \vec{b}_{f_{ib}}^b - (2\vec{\omega}_{ie}^e) \times \vec{v}_{eb}^e + \vec{g}^e}_{F\vec{x}} + \underbrace{C_b^e \vec{f}_{ib,meas}^b}_{G_u \vec{u}} - \underbrace{C_b^e \vec{w}_{accel}}_{G_w \vec{w}} + \underbrace{\vec{f}_{\dot{v}_{eb}}^e}_{\dot{\vec{x}}} \quad (17)$$

Hence, an EKF state vector can be formulated and generalized under the IMU fault conditions as (18) in a continuous form,

$$\dot{\vec{x}}' = F\vec{x} + G_u \vec{u} + G_w \vec{w} + \vec{f}_{\dot{\vec{x}}} \quad (18)$$

where $\vec{f}_{\dot{\vec{x}}}$ is a fault vector on a state vector due to IMU sensor faults in a continuous form. And a discrete form of (18) is shown in (19).

$$\vec{x}_k' = \Phi_k \vec{x}_{k-1} + \Gamma_{uk} \vec{u}_k + \vec{w}_k + \vec{f}_{\vec{x}_k} \quad (19)$$

3.1.2 Propagation of an IMU fault impact through the EKF

The discrete form of a faulty state vector in (19) is propagated through the EKF state propagation equations as in (1) and (2). To simulate the propagated IMU fault impacts, a nominal state vector $\bar{\mathbf{x}}_k$ is replaced with the faulty state vector $\bar{\mathbf{x}}_k'$ derived in (19). A predicted state at time k becomes,

$$\bar{\mathbf{x}}_k' = \Phi_k \hat{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k} \quad (20)$$

Substituting (20) into (2) yields a measurement updated state as

$$\begin{aligned} \hat{\mathbf{x}}_k' &= \bar{\mathbf{x}}_k' + K_k' (\bar{\mathbf{z}}_k - H_k \bar{\mathbf{x}}_k') \\ &= (I - K_k' H_k) (\Phi_k \hat{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k}) + K_k' \bar{\mathbf{z}}_k \end{aligned} \quad (21).$$

Here, K_k' represents the Kalman gain matrix computed by the faulty state transition matrix, due to erroneous attitude information by the faulted IMU measurements. For simplicity, we introduce a new matrix $L_k' = (I - K_k' H_k)$. Subsequently, (21) is simplified as (22):

$$\begin{aligned} \hat{\mathbf{x}}_k' &= L_k' (\Phi_k \hat{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k}) + K_k' \bar{\mathbf{z}}_k \\ &= L_k' (\Phi_k \hat{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k) + K_k' \bar{\mathbf{z}}_k + L_k' \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k} \end{aligned} \quad (22)$$

Based on the derived expression in (22), it is noticeable that the vector $L_k' \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k}$ is additionally added due to the IMU fault measurements at time k.

3.1.3 Derivation of an EKF state error due to an IMU fault

A VPL against an IMU sensor fault should be formulated to fully overbound a state error to the level of integrity. Thus, the state error equation is derived as $\tilde{\mathbf{x}}_k = \hat{\mathbf{x}}_k' - \bar{\mathbf{x}}_k$, where $\bar{\mathbf{x}}_k = \Phi_k \bar{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\omega}_k$ represents a true state. The steps for deriving the state error equation are then:

- 1) Multiply L_k' to $\bar{\mathbf{x}}_k$;

$$L_k' \bar{\mathbf{x}}_k = L_k' \Phi_k \bar{\mathbf{x}}_{k-1} + L_k' \Gamma_{uk} \bar{\mathbf{u}}_k + L_k' \bar{\omega}_k \quad (23)$$

- 2) Subtract $L_k' \bar{\mathbf{x}}_k$ from $\hat{\mathbf{x}}_k'$;

$$\hat{\mathbf{x}}_k' - L_k' \bar{\mathbf{x}}_k = L_k' (\Phi_k \hat{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k}) + K_k' \bar{\mathbf{z}}_k - L_k' \bar{\mathbf{x}}_k \quad (24)$$

- 3) Insert $\bar{\mathbf{z}}_k = H_k \bar{\mathbf{x}}_k + \bar{\mathbf{v}}_k$ to (24) where $\bar{\mathbf{v}}_k$ is a measurement noise vector;

$$\hat{\mathbf{x}}_k' - L_k' \bar{\mathbf{x}}_k = L_k' (\Phi_k \hat{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k}) + K_k' (H_k \bar{\mathbf{x}}_k + \bar{\mathbf{v}}_k) - L_k' (\Phi_k \bar{\mathbf{x}}_{k-1} + \Gamma_{uk} \bar{\mathbf{u}}_k + \bar{\omega}_k) \quad (25)$$

- 4) Rearrange (25)

$$\tilde{\mathbf{x}}_k = L_k' \Phi_k \tilde{\mathbf{x}}_{k-1} + L_k' \bar{\mathbf{f}}_{\bar{\mathbf{x}}_k} + K_k' \bar{\mathbf{v}}_k - L_k' \bar{\omega}_k \quad (26)$$

A state error at time k can be expressed as a function of a previous state vector, a state fault vector, a measurement noise vector, and a process noise vector as in (26). The state error equation indicates that the fault impact on the current fault vector due to IMU sensor faults recursively affects the derived current time state error vector.

3.2 Error bounding under IMU fault conditions

3.2.1 EKF innovation

An EKF innovation in a GNSS/IMU integrated system is defined as the difference between the GNSS pseudorange measurements and the IMU sensor predicted range. The IMU sensor predicted range is determined using a position estimate computed by IMU sensor measurements and the GNSS satellite position information. (27) shows an analytical expression of an EKF innovation to investigate the impact of the IMU sensor fault on an EKF innovation.

$$\begin{aligned}\vec{\gamma}_k &= \vec{z}_k - H_k \vec{x}_k \\ &= H_k (\Phi_k \vec{x}_{k-1} + \Gamma_{uk} \vec{u}_k + \vec{\omega}_k) + \vec{v}_k - H_k (\Phi_k \hat{x}_{k-1} + \Gamma_{uk} \vec{u}_k + \vec{f}_{\vec{x}_k}) \quad (27) \\ &= -H_k \Phi_k \tilde{x}_{k-1} - H_k \vec{f}_{\vec{x}_k} + H_k \vec{\omega}_k + \vec{v}_k\end{aligned}$$

It is observed that an EKF innovation and the state error vector include the state fault vector introduced in (19) in an identical form. This finding motivates the utilization of an EKF innovation for bounding the state error derived in (26) in real time. The following section introduces the derivation of an EKF innovation-based VPL against the IMU sensor fault hypothesis.

3.2.2 Derivation of EKF innovation-based VPLs

In Section 3.1.3 and Section 3.2.1, the expressions for a state error vector and an EKF innovation due to faulty IMU sensor measurements are derived. As stated in the previously, both expressions contain the state fault vector. To simplify the equations, a new matrix A_k is introduced as in (28), which is a common term in both the state error and innovation, and is an impact of the IMU sensor faults in the state domain. Subsequently, (26) and (27) are represented as a function of A_k as shown in (29) and (30).

$$A_k = \Phi_k \tilde{x}_{k-1} + \vec{f}_{\vec{x}_k} \quad (28)$$

$$\tilde{x}_k = L_k' A_k + K_k' \vec{v}_k - L_k' \vec{\omega}_k \quad (29)$$

$$\vec{\gamma}_k = -H_k A_k + H_k \vec{\omega}_k + \vec{v}_k \quad (30)$$

This study exploits the relationship between the state error and EKF innovation to derive a real-time VPL equation. Without knowing the state fault vector, determining A_k directly is not possible. Instead, we found that A_k can be obtained using the EKF innovation $\vec{\gamma}_k$ based on (30). (31) represents the derived $A_{k,p}$ using the EKF innovation. Here, the subscript p on $A_{k,p}$ means the partial matrix of A_k . Because an EKF does not take measurements for all the EKF states, $(H_k^T H_k)$ becomes a singular matrix, and is not invertible. Thus, only non-zero terms of H_k matrix is used to compute the matrix $A_{k,p}$ as follows. The resulting $A_{k,p}$ matrix can capture a position state bias from the EKF innovation vector:

$$A_{k,p} = -(H_{k,p}^T H_{k,p})^{-1} H_{k,p}^T (\vec{\gamma}_k - \vec{v}_k - H_{k,p} \vec{\omega}_{k,p}) \quad (31)$$

By inserting $A_{k,p}$ into (29), the position state error can be obtained using an innovation vector, which we can access in the EKF in real time.

$$\begin{aligned}\tilde{x}_{k,p} &= -L_{k,p}' (H_{k,p}^T H_{k,p})^{-1} H_{k,p}^T (\vec{\gamma}_k - \vec{v}_k - H_{k,p} \vec{\omega}_{k,p}) + K_{k,p}' \vec{v}_k - L_{k,p}' \vec{\omega}_{k,p} \\ &= -L_{k,p}' (H_{k,p}^T H_{k,p})^{-1} H_{k,p}^T \vec{\gamma}_k + L_{k,p}' (H_{k,p}^T H_{k,p})^{-1} H_{k,p}^T \vec{v}_k + K_{k,p}' \vec{v}_k\end{aligned} \quad (32)$$

For simplicity, if we introduce a matrix B_k in (33), subsequently the state error equation is simplified as (34).

As users have no information about the noise, the noise terms should be properly bounded within the VPL equations. Based on the state error expression as a function of EKF innovation, a VPL against IMU sensor faults is formulated as (35), which includes the position state bias estimates and uncertainty noise bounding terms as in (35). The uncertainty noise bounding terms include uncertainty on the bias estimates ($K_{md,IMU}\sigma_{B_k\vec{v}_k,v}$) and uncertainty on the remaining position error ($K_{md,IMU}\sigma_{K_k\vec{v}_k,v}$). $K_{md,IMU}$ is applied to assure that the derived VPLs against IMU faults bound the user position to the level of probability that is allocated to the IMU fault hypothesis. As the bias estimates are computed in real time, it is possible to have PLs surrounding the bias estimates, two for each side. This results in a lower level of conservatism compared to using an absolute VPL.

$$B_k = L'_{k,p} (H_{k,p}^T H_{k,p})^{-1} H_{k,p}^T \quad (33)$$

$$\begin{aligned} \tilde{x}_{k,p} &= -B_k \vec{\gamma}_k + B_k \vec{v}_k + K'_k \vec{v}_k \\ &= -B_k \vec{\gamma}_k + (B_k + K'_k) \vec{v}_k \end{aligned} \quad (34)$$

$$VPL_{IMU,KF} = -B_k \vec{\gamma}_k \pm \left\{ K_{md,IMU} \sigma_{B_k \vec{v}_k, v} + K_{md,IMU} \sigma_{K_k \vec{v}_k, v} \right\} \quad (35)$$

Fig. 3 shows an example of the $VPL_{IMU,KF}$ based on the derived equation. In this simulation, a step IMU fault vector, which has magnitudes of 1.5 m/s^2 on the accelerometer and 0.03 rad/s on the gyroscope, is injected to all three axes both for accelerometer and gyroscope for 10 seconds after 2 minutes of operation. The vertical position errors shown in the dotted curve are determined by the difference between the EKF estimated vertical positions and the true vertical positions. The blue solid curves indicate the $VPL_{IMU,KF}$ s. It is shown that $VPL_{IMU,KF}$ s determined by innovation vectors well bound the state errors. In this example, the K value of the missed detection probability of the IMU is chosen to be 3.66, which bounds the noise of the state bias estimates with a missed detection probability of around $0.25 \cdot 10^{-3}$. That is, the two real time VPLs can overbound the true vertical position state errors with this probability of missed detection during UAV operations.

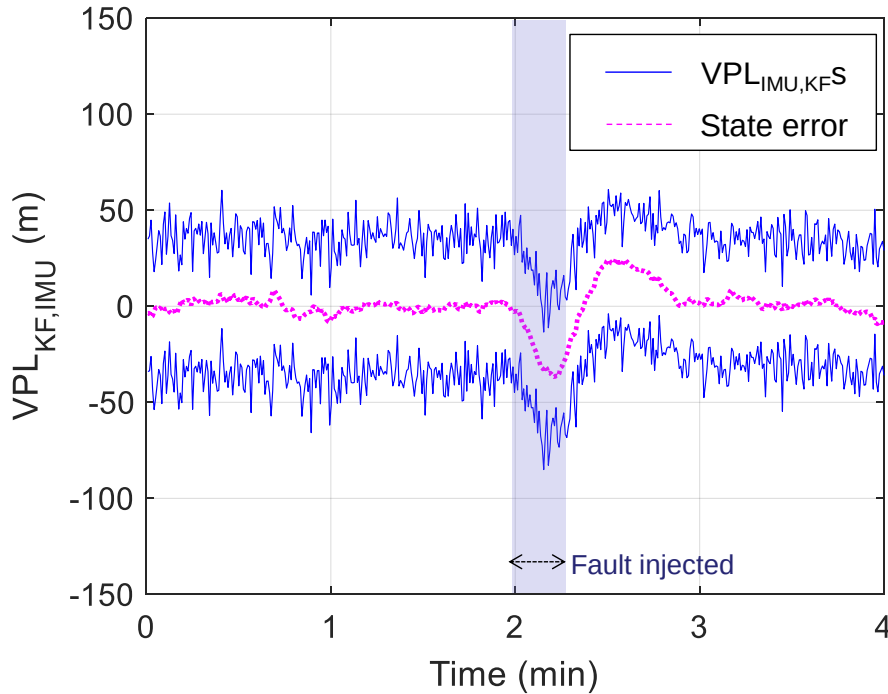


Fig. 3. An example simulation results of derived VPL equations

4.0 SIMULATION RESULTS UNDER ASPECIFIC IMU-FAULT CONDITION

4.1 Simulation conditions

Simulations are conducted to derive VPLs for UAVs in flight for each fault hypothesis. For these simulations, sensor noise for GNSS measurements and IMU measurements are generated, and corresponding noise parameters are also applied to the KF to bound the output KF variance. This section introduces the noise parameters as well as additional simulation conditions used for generating VPLs.

4.1.1. UAV flight path

This simulation is based on a specific UAV flight path configured as a combination of a straight phase and a landing phase, as shown in Fig. 4. The simulation period is 4 minutes, including a 3-minute straight phase and a one-minute landing phase. A user is assumed to be located in Daejeon, South Korea for the simulation.

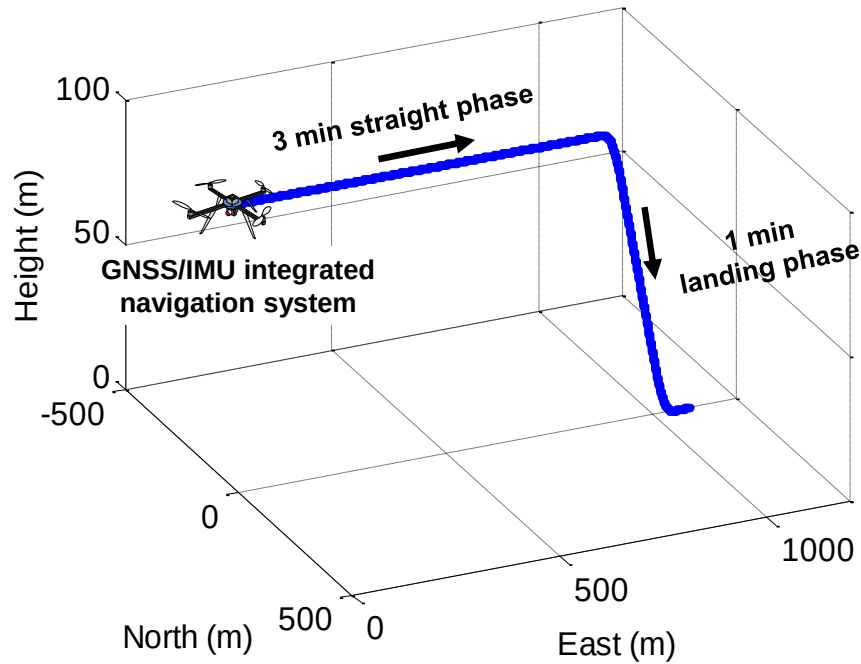


Fig. 4. UAV flight path used for the simulation studies

4.1.2. Sensor noise levels

For the simulation, a simulated 31-satellite GPS constellation using GPS orbit parameters is used. Table 1 shows the standalone GNSS sensor noise levels used for bounding KF output variance. To bound the KF output covariance, bounded variances for each error source are applied within the KF.

Table 1. Standalone GNSS noise levels used for the simulation

Standalone GNSS (31-satellite GPS constellation)			
SIS error [21]	Ionospheric error [22]	Tropospheric error [2]	Multipath error [3]
URA 2.4 m	Residual ionospheric sigma after correction	WAAS MOPS model	AAD-A model

To describe a low-cost UAV hardware configuration, a consumer-grade IMU is used for the simulation with the parameters shown in Table 2.

Table 2. A consumer grade IMU noise level used for the simulation [14]

IMU sensor (Consumer-grade IMU)			
Accelerometer		Gyroscope	
Bias PSD (m^2/s^5)	Noise PSD (m^2/s^3)	Bias PSD (rad^2/s^3)	Noise PSD (rad^2/s)
10^{-5}	0.2^2	$4 \cdot 10^{-11}$	0.01^2

4.1.3. Integrity risk allocation

For the simulations of VPL_{KF} s, the total system integrity requirement in the vertical domain were set as $P_{\text{HMI},V} = 10^{-6}$, and the prior probabilities of a GPS sensor fault and an IMU sensor fault were assumed to be $P_{\text{prior,GPS}} = 10^{-4}$ and $P_{\text{prior,IMU}} = 10^{-3}$, respectively, considering the hardware qualities used for low-cost UAVs. Further, the total integrity probability was allocated 50% to the nominal hypothesis, 25 % to the GNSS fault hypothesis, and 25% to the IMU sensor fault hypothesis (i.e., $0.5 P_{\text{HMI},V}$, $0.25 P_{\text{HMI},V}$, and $0.25 P_{\text{HMI},V}$, respectively). However, for multi-sensor systems, it is not ideal to allocate the total system integrity to each fault hypothesis in advance because the type of sensor used and the sensor quality would be different for each system. Thus, this study proposes to optimally allocate the total system integrity requirements in real time. The concept of optimal integrity allocation has been proposed for the RAIM architecture [4], and this study adopted the concept for the UAV navigation systems. The core idea of the optimal integrity allocation is to allocate the total system integrity to make the final VPL_{KF} smallest while applying a constraint where the sum of the integrity allocations should be the same as the total system integrity. A VPL_{KF} determined by the optimal integrity allocation would provide equal or smaller VPLs compared to the VPLs for each fault hypothesis computed by fixed integrity allocations. For more details on the optimal allocation concept, see [4].

4.2 Simulation under an IMU-fault condition

Fig. 5. shows a simulation result conducted for 4 minutes under a partial GNSS constellation. The figure shows the vertical position state error and corresponding VPL_{KF} for each fault hypothesis under the same step IMU fault case applied in Fig. 3. Specifically, a step IMU fault which has magnitudes of 1.5 m/s^2 on the accelerometer and 0.03 rad/s for the gyroscope is injected to all three axes both for accelerometer and gyroscope for 10 seconds after 2 minutes of operation. The magnitude of the fault was chosen arbitrary within a reasonable range for the example simulation. Of course, many different fault types and magnitudes could occur, and these will be investigated in future work.

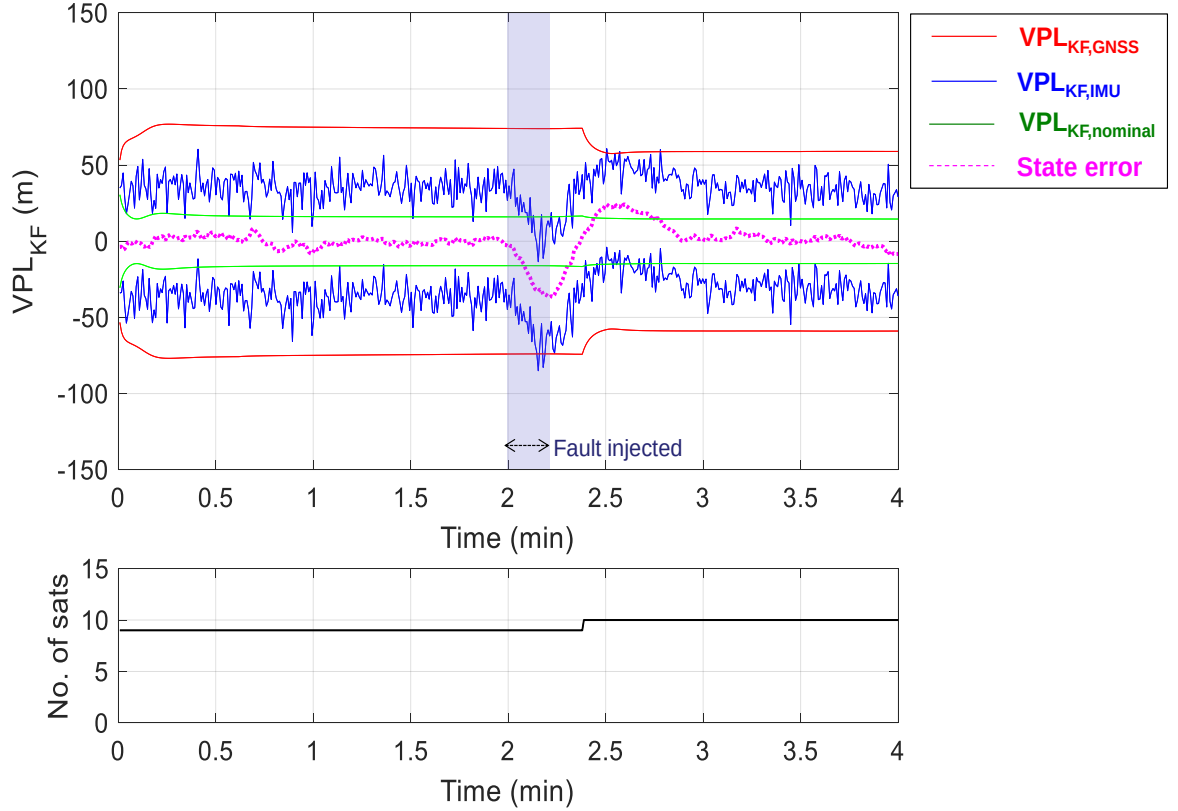


Fig. 5. Vertical position state error and corresponding VPL_{KF} for each fault hypothesis under a step IMU fault case injected for 10 seconds after 2 minutes of operation (top); number of satellites (bottom)

In the upper plot of Fig. 5, three VPL_{KFs} are shown for each fault hypothesis as well as state errors. It is shown that the $VPL_{KF,IMU}$ well bounds the state error (the pink dashed line) under these IMU fault conditions with two upper and lower $VPL_{KF,IMU}$ s. By taking advantage of the estimated fault bias from the innovation vector, asymmetric $VPL_{KF,IMU}$ values are created around the estimated fault bias.

VPL s for nominal and GNSS fault conditions are also plotted in Fig. 5, and the number of GPS satellites used by the KF is shown in the lower plot of Fig. 5. $VPL_{KF,GNSS}$ is highly dependent on satellite geometry. In Section 5.0, this simulation is extended over 24 hours to cover all satellite geometries under no-IMU fault conditions (to simulate typical performance in most situations).

5.0 SIMULATION UNDER NO-IMU-FAULT CONDITION

This section introduces $VPL_{KF,opt}$ simulated under no-IMU-fault conditions to see the typical performance of GNSS/IMU integrated navigation system with the parameters assumed above. To cover the 24-hour ground track repeat time of the GPS constellation, the 4 minute UAV flight simulation shown in Fig. 4 was independently conducted 360 times both for 31-satellite L1 and L1/L5 frequency standalone GPS/IMU integrated systems.

The results of this simulation are shown in Fig. 6. Due to the conservatism of the WLS-RAIM algorithm on computing $VPL_{KF,GNSS}$, $VPL_{KF,GNSS}$ dominates the overall $VPL_{KF,opt}$ under no-IMU-fault conditions. The blue curve shows the results for the L1 standalone GPS/IMU case, and the orange curve shows the results for the L1/L5 dual-frequency standalone GPS/IMU case (in which ionosphere-free pseudoranges are used). The rapid changes that appear in these two curves are an artifact of re-starting the UAV flight simulation (and re-initializing the KF) every 4 minutes.

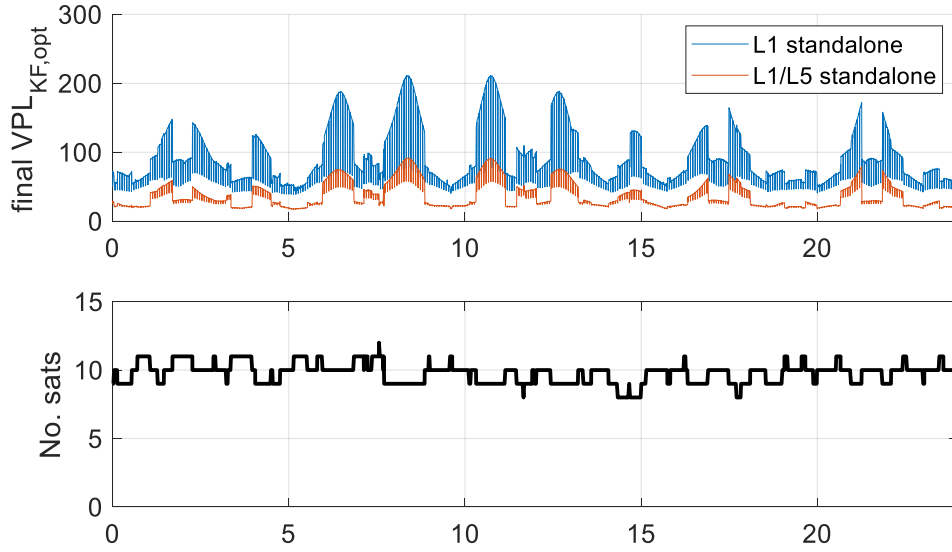


Fig. 6. Final $VPL_{KF,opt}$ of a GNSS/IMU integrated system for L1 single frequency standalone GPS/IMU (blue curve), and L1/L5 dual frequency GPS/IMU (orange curve) (top); number of satellites for 24 hours (bottom)

Based on the $VPL_{KF,opt}$ results, levels of achievable $VPL_{KF,opt}$ as functions of achievable availability levels are analyzed in Fig. 7 both for an L1 standalone GPS/IMU system and an L1/L5 standalone GPS/IMU system. Fig. 6 and Fig. 7 clearly show the superiority of the dual-frequency L1/L5 approach and its capability of achieving high availability of VPLs less than 100 meters.

In this study, an example result was obtained under a specific UAV flight trajectory and a specific grade of IMU sensor noise level. However, these analysis results under various conditions can be utilized to propose an operational concept for UAVs. The achievable level of $VPL_{KF,opt}$ can be utilized for determining the safety margin when determining the allowable flight region for low-flying UAVs constrained by an upper altitude limit and a minimum height above obstacles and terrain, as shown in Fig. 8.

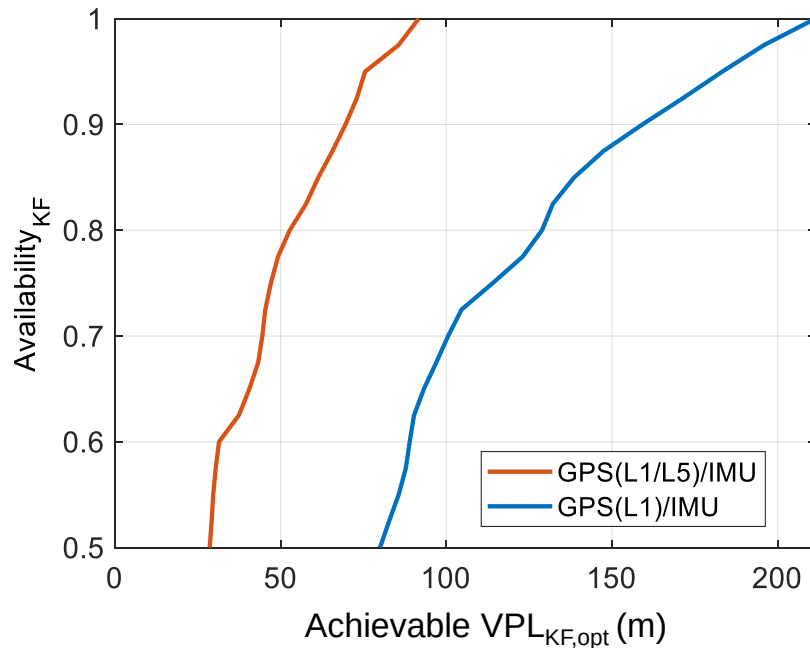


Fig. 7. Achievable $VPL_{KF,opt}$ as a function of availability for L1 standalone GPS/IMU integrated system (blue curve) and L1/L5 standalone GPS/IMU integrated system (orange curve)

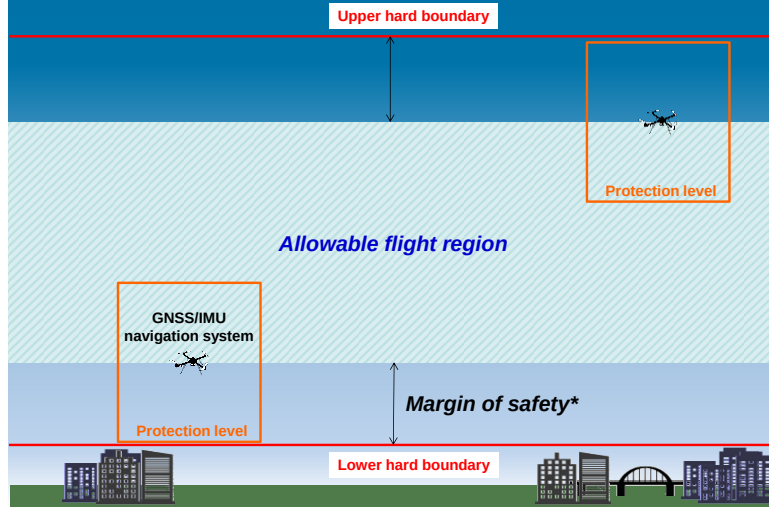


Fig. 8. UAV operation concept. The margin of safety can be determined based on the achievable VPL_{KF} .

In multi-sensor systems, the type of sensor used and the sensor quality will differ over various systems and applications. Thus, multi-dimensional analyses for the worst $VPL_{KF,opt}$ s for all simulation times were performed by changing three probabilities: the total HMI probability in the vertical domain, the prior probability of a GPS sensor fault, and the prior probability of an IMU sensor fault. In Fig. 9, the x- and y-axes indicate the prior probabilities of these sensor faults, and the z-axis indicates the total system integrity requirement. For this analysis, L1/L5 dual-frequency-based standalone GPS and a consumer grade IMU are used. Note that the VPLs represented by colors in Fig. 9 represent the worst values observed over 360 repetitions of the 4-minute UAV flight simulation, as shown in Fig. 6.

The results in Fig. 9 indicate that as the prior probabilities of sensor fault increase, and the total integrity requirement becomes tighter, the resulting maximum $VPL_{KF,opt}$ s increases. $VPL_{KF,opt}$ is more sensitive to increasing GPS fault probability than it is to increasing IMU fault probability because the VPL under the GPS failure hypothesis ($VPL_{KF,GNSS}$) tends to dominate the real-time results due to the conservatism of WLS RAIM, as explained above. This analysis result can be used to determine the achievable VPL performance with given probabilities or to provide information to determine the required navigation sensor qualities of UAVs under given system integrity requirements.

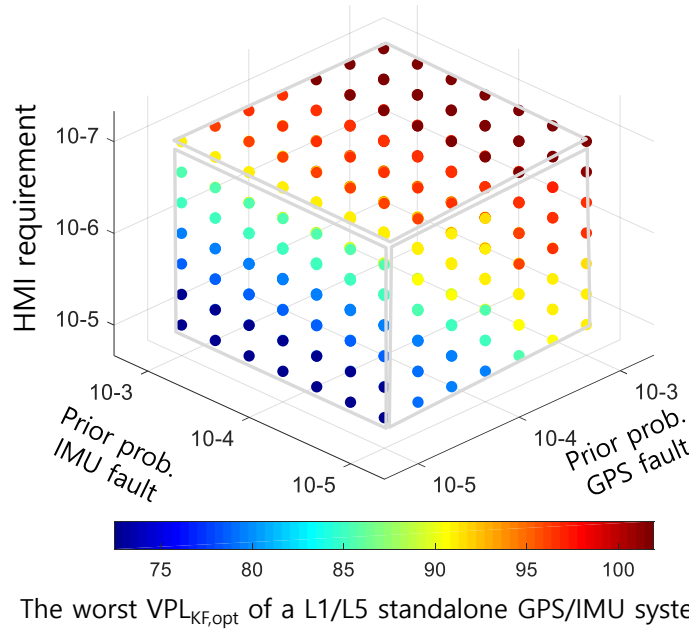


Fig. 9. Multi-dimensional analysis of the worst $VPL_{KF,opt}$ as functions of a total HMI probability in the vertical domain, prior probability of a GPS sensor fault, and prior probability of an IMU sensor fault.

6.0 CONCLUSIONS

This study proposed an integrity architecture for a GNSS/IMU integrated system. EKF VPLs were derived for each sensor fault hypothesis, including the IMU fault hypothesis to provide complete integrity assurance for integrated UAV navigation systems. EKF innovations provided direct estimates of user position bias under IMU fault conditions, removing conservatism. Kalman-filter based VPLs were simulated for a specific flight scenario based on a standalone GPS and a consumer grade IMU integrated system. Simulation results showed that the derived VPL under an IMU step fault hypothesis successfully assures user position error under IMU fault conditions. Simulation results under no-IMU-fault condition indicated that the largest VPL_{KF} is overbounded by 105 m for L1/L5 dual-frequency GPS case under most situations (no IMU fault conditions) for every combination of the total probability of HMI and the prior probability of sensor faults within a given range (as shown in Fig. 9).

These analysis results can be utilized to suggest an operational concept for UAVs flying within the designed airspace, as shown in shown in Fig. 8. In this concept, an achievable VPL_{KF} for a given availability requirement from the nominal simulation results is chosen as VAL, and this VAL is continuously compared with the real-time VPL_{KFS} . These VALs are also used to define the allowed UAV flight region (the shaded area in Fig. 8) so that UAVs do not violate the upper and lower restricted areas due to navigation errors or navigation faults. As the VPLs that UAV integrated navigation systems can achieve get smaller due to higher-quality sensors or relaxed integrity requirements, VALs could also decrease, and the allowed flight region could be enlarged.

This study is a stepping stone for research on more-detailed KF-based multi-sensor integrity systems. More rigorous studies, mitigating worst-case assumptions on WLS RAIM by employing GNSS augmentation systems, including correlated IMU noise modeling in a recursive estimator for bounding the KF covariance, analyses under different and more-challenging flight conditions, and integrity assurance under different sensor integration scenarios will be studied in future work.

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